

- Goal: Construct predictive intervals for individual treatment effects with finite-sample validity and competitive efficiency.
- Challenge: The individual treatment effect is unobserved (only one potential outcome is seen), complicating calibration.
- Existing meta-learners provide points, not distribution-free intervals for individual treatment effects.

- Bayesian models are architecture-specific; frequentist coverage can fail under misspecification.
- Weighted conformal prediction infers on nuisance parameters (potential outcomes), not directly on individual treatment effects.
- Covariate shift between pseudo-targets and true individual treatment effects can break naive conformal calibration.

- Layer standard split conformal prediction on top of two-stage pseudo-outcome meta-learners.
- Calibrate directly on pseudo-outcomes designed to mimic the individual treatment effect, preserving conformal validity.
- Model-agnostic: works with any machine learning architecture for nuisance and treatment-effect stages.

- Three-way data split: nuisance fitting, treatment-effect training on pseudo-outcomes, and conformal calibration.
- Construct pseudo-outcomes via IPW, X-, or DR learners; then fit an individual treatment effect predictor to these pseudo-outcomes.
- Use conformity scores on a held-out split of pseudo-outcomes to build intervals.

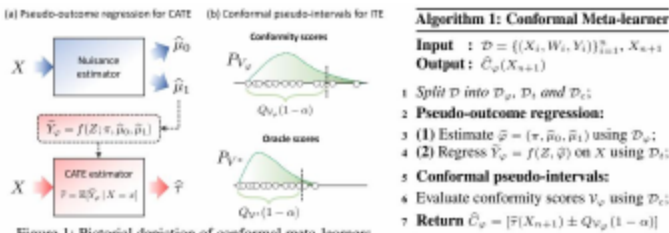


Figure: Pictorial depiction of conformal meta-learners.

- Split the data into: a nuisance set for outcome models for control and treatment and the (known) propensity; a training set for the treatment-effect predictor using pseudo-outcomes; and a calibration set for conformal prediction.
- Fit nuisance models for outcome regression (and use the (known) propensity when assumed).
- Train any regressor for the individual treatment effect on features and pseudo-outcomes.
- Compute conformity scores on the calibration split and take empirical quantiles for interval construction.

- Conformity score: absolute error between the pseudo-outcome and the prediction; signed-distance variants are possible with quantile regressors.
- The interval width is set using a calibration quantile: take the empirical quantile at the coverage level computed as one minus alpha, adjusted by the usual finite-sample factor based on the size of the calibration split.

$$[\hat{\tau}(x) \pm q_{\alpha}]$$

- IPW: Reweights observed outcomes to debias treatment assignment.
- X: Imputes missing potential outcomes then differences them.
- DR: Combines reweighting with outcome regression to reduce variance.

- Compare conformity scores on pseudo-outcomes versus unobserved oracle individual-treatment-effect scores.
- If pseudo-outcome scores stochastically dominate oracle scores, conformal coverage transfers to the true target.
- Analyze via first-order, second-order, and monotone-convex dominance.

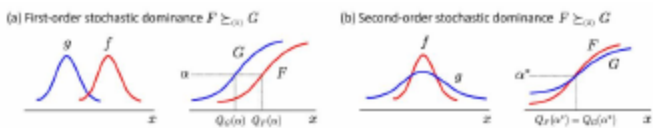


Figure: Illustration of stochastic dominance between two distributions.

- First-order dominance: marginal validity across all confidence levels.
- Second-order or monotone-convex dominance: valid for practically relevant high coverages (for example, alpha up to 0.1).
- Links pseudo-outcome design to achievable coverage levels.

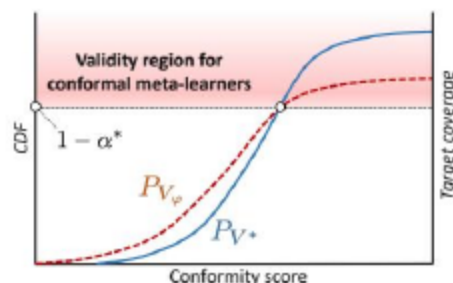


Figure: Typical target coverage levels; alpha commonly set to 0.05 or 0.1.

- With known propensity and absolute-residual scores, IPW and DR satisfy monotone-convex dominance in a model- and distribution-free manner.
- The X-learner admits no distribution-free stochastic order and may undercover.
- With signed-distance scores, the order reverses for IPW and DR; use with care.

- Model-free guarantees rely on known propensity; misspecification can harm dominance.
- The X-learner may undercover when imputation errors are asymmetric.
- The maximal valid confidence level is data-dependent; estimating it remains challenging.

- With known propensity and absolute-residual scores: IPW and DR satisfy monotone-convex dominance; X has no distribution-free order.
- With signed-distance scores: the dominance direction reverses for IPW and DR; apply with caution.

- Architecture-agnostic: plug in trees, boosted models, or deep networks for the treatment-effect predictor.
- Calibrates on pseudo-outcomes paired with X drawn from the test-like distribution, mitigating covariate shift.
- Limitations: model-free guarantees require known propensity; tight confidence levels are data-dependent.

- Settings: synthetic; semi-synthetic IHDP and NLSM; ACIC2016 (77 tasks).
- Baselines: weighted conformal variants (naïve, exact, inexact), IPW, X, and DR variants.
- Metrics: empirical coverage, average interval length (efficiency), and CATE RMSE.

Synthetic Results Summary

- Empirical CDFs show DR/IPW pseudo-score dominance over oracle in many settings, predicting validity.
- At 90% target coverage: DR attains coverage with shorter intervals than valid weighted-conformal baselines.
- The X-learner yields low RMSE but undercovers due to lack of dominance.

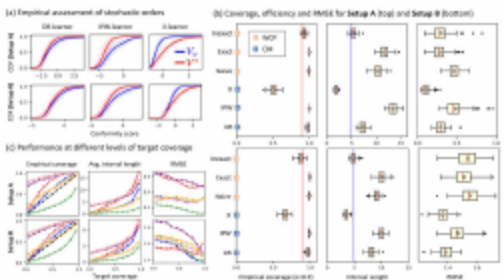


Figure: Performance in the synthetic setting; vertical lines denote target coverage and optimal width; color encodes methods.

Semi-synthetic Dominance

- On IHDP/NLSM, DR/IPW pseudo-score CDFs typically lie below oracle (first-order dominance).
- The X-learner is often dominated by oracle scores, leading to undercoverage.
- Dominance patterns align with observed coverage and interval efficiency.

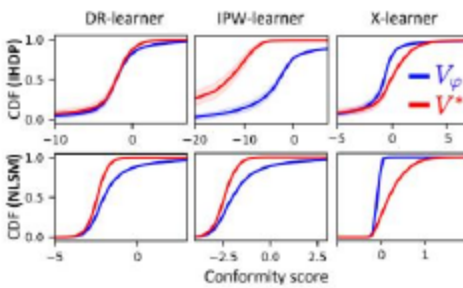


Figure: Stochastic orders in IHDP/NLSM.

- DR achieves near-target or better coverage with shorter intervals than valid weighted-conformal methods.
- IPW is valid but with very wide intervals and high RMSE on IHDP.
- X undercovers despite strong RMSE, consistent with theory.

Table: Performance of all baselines in semi-synthetic datasets. Bold numbers correspond to best performance.

Method	IHDP			NLSM		
	Coverage	Avg. len.	RMSE	Coverage	Avg. len.	RMSE
Naïve	0.89 (0.02)	18.9 (4.04)	4.73 (1.00)	0.99 (0.00)	4.82 (0.02)	0.15 (0.00)
Exact	0.99 (0.00)	29.8 (7.60)	4.50 (0.97)	0.99 (0.00)	4.92 (0.07)	0.19 (0.01)
Inexact	0.61 (0.04)	8.49 (1.36)	4.61 (0.99)	0.96 (0.00)	2.38 (0.01)	0.18 (0.00)
X	0.65 (0.04)	11.0 (3.04)	3.34 (0.56)	0.27 (0.00)	0.38 (0.00)	0.14 (0.00)
IPW	0.99 (0.00)	112 (23.0)	19.9 (3.44)	0.99 (0.00)	6.48 (0.07)	0.61 (0.02)
DR	0.96 (0.01)	16.7 (3.30)	3.32 (0.53)	0.99 (0.00)	6.24 (0.07)	0.37 (0.01)

- Valid methods consistently meet or exceed 0.9 coverage targets.
- DR combines short valid intervals with low RMSE across diverse settings.
- The X-learner undercovers, reflecting lack of distribution-free dominance.



Figure: Performance across 77 ACIC tasks.

- DR pseudo-outcomes reduce variance via augmentation, stabilizing conformity scores.
- Empirical dominance often widens the valid confidence range beyond conservative guarantees in practice.
- Calibrating at the individual-treatment-effect level avoids compounding errors from nuisance-level conformal steps.

- A general, model-agnostic framework to produce predictive intervals for individual treatment effects via conformal meta-learners.
- Stochastic ordering theory linking pseudo-outcome design to coverage validity.
- Empirically, DR-based conformal meta-learners achieve valid coverage with competitive efficiency and strong accuracy.

- Extend guarantees to unknown propensity via robust estimation or sensitivity bounds.
- Design new pseudo-outcome transformations optimizing efficiency under dominance.
- Leverage modern architectures within the same calibration framework for high-dimensional covariates.

- Thank you for your attention!
- Questions and feedback are welcome.

Acknowledgments

- We thank collaborators, institutions, and funding bodies that supported this research.
- Gratitude to the community for datasets and evaluation benchmarks (IHDP, NLSM, ACIC).